

# B02 Gate B Protected Review - E-commerce Returns

Protected diagnostic review - no answer leakage to later Gates

## Protected review - saved genuine Gate attempt required

Review the saved attempt, diagnose the weak control, close this PDF, complete targeted repair, then save a changed micro-proof. Gate B/C exposure still requires the existing explicit reassignment rule.

### Objective fixture controls

| Check                           | Protected control             |
|---------------------------------|-------------------------------|
| Raw rows                        | 120                           |
| Valid numeric rows              | 119                           |
| Malformed numeric               | record 5038 - refund_unit=BAD |
| Master rows / unique keys       | 31 / 30                       |
| Duplicate master key            | S013                          |
| Valid-row control total         | 93090.00                      |
| Naive dirty-master merge rows   | 121                           |
| Naive dirty-master merged total | 93530.00                      |
| Duplicate-key inflation         | 440.00                        |

### Interpretation - why a plausible pipeline can still be wrong

- Record 5038 contains a deliberately malformed numeric. If code uses errors='coerce' and silently drops the resulting null, it can manufacture a clean-looking output while hiding a source defect.
- S013 appears twice in the master. 2 valid fact rows hit that key; a naive merge inflates the control by 440.00.
- All fact keys are present in the master, so an unmatched-key count of zero does NOT prove the merge is safe. Uniqueness/cardinality must be checked separately.
- A critical validation failure means the trusted publish path stays closed. Logs and a manifest should show the failed run and the withheld output.

## Reasoning and evidence rubric

- File contract: select exactly the three intended domain files in deterministic order and record source lineage.
- Schema/numeric contract: validate required columns/types; identify the malformed value explicitly; preserve evidence rather than hiding it.
- Master contract: assert uniqueness BEFORE merge. Do not pick one duplicate master row without an approved business rule.
- Merge contract: declare many-to-one and reconcile pre/post row count plus control totals. Explain any row multiplication before aggregating.
- Validation gate: critical FAIL prevents final trusted output. A warning-only publish for a critical defect is a critical failure.
- Run evidence: log selected files, counts, checks, runtime/library versions, output status and failure/recovery path.
- Rerun evidence: a second run must not double append or silently create a second competing trusted output.
- Defense: explain input grain, output grain, why the checks are placed before publish, and what Course 3 adds later.

### Critical failures

- Silent coercion/drop/dedupe that hides the injected defect.
- Merge before proving master-key uniqueness.
- Publishing a trusted output after a critical validation FAIL.
- No saved failure/recovery evidence or unsafe second-run duplication.
- Unable to explain grain, merge cardinality or validation/rerun behavior.

## **Repair and changed retry route**

- Save the original failed/developing Gate attempt. Do not overwrite it with the repaired version.
- Open this matching protected review only after that saved attempt exists.
- Repair only the named weak competency using the targeted remediation material.
- Close this review. Produce a changed micro-proof on a different small file/key/failure condition so the repaired skill is demonstrated without copying the protected controls.
- Save the changed micro-proof plus updated validation evidence. Only then may the existing Mentor state machine allow an explicit fresh full-retest assignment when required.

### **Next route**

PASS: B02 ends. FAIL/developing: second targeted repair; Gate C remains sealed until the full evidence chain and explicit reassignment are satisfied.